

SOLPS-ITER v. 3.0.6 Release Notes

These Release Notes detail changes between SOLPS-ITER versions 3.0.5 and 3.0.6 and give information about how to transition an older SOLPS-ITER run to be continued with the 3.0.6 code version. Users are urged to read this document carefully and thoroughly before proceeding with use of SOLPS-ITER 3.0.6. Users should also consult the updated version of the SOLPS-ITER manual that accompanies this distribution and other documents posted to the SOLPS-ITER web site. For full details of all updates, please consult the GIT repository logs. In addition to the items listed below, the new code version also contains lots of bug fixes, new documentation, and output clarification/prettifying improvements, as reported/requested by users and developers over the last few months.

1. How to obtain SOLPS-ITER v.3.0.6:

In order to update your copy of the SOLPS-ITER code suite from version 3.0.5 to 3.0.6, use the *"solps-iter_update"* script. Then run *"switch_to_master"*. As of version 3.0.6, the SOLPS-ITER distribution will, by default, point to the master branch of the SOLPS-ITER GIT repository and its submodules. If you wish to point instead to the develop branch, then use the *"solps-iter_update_develop"* script after the first two. As usual, before making any such update, please make sure that any local changes you might have are committed to local GIT branches or stashed, so as to avoid them being overwritten.

Users that also need the SOLPS4-5 converter program *b2sxdx* should update their copy of SOLPS4.3 as well, as some bug fixes were introduced during the SOLPS-ITER 3.0.6 testing process.

For subsequent updates, users should use *"solps-iter_update"* or *"solps-iter_update_develop"*, depending on which branch they are using, and only once per update.

SOLPS-ITER v.3.0.6 combines work that has proceeded in the *feature/averaging*, *feature/IDS*, *feature/ammonia* and *feature/new_friction* branches of the repository, and which should all now be considered obsolete (with the possible exception of *feature/IDS*, where development continues). The obsolete branches will soon be removed from the reference repository and users should no longer point to them.

Because the update to v.3.0.6 involves the setting of new environment variables, loading of new modules (in particular, if you want IMAS support, you should point to *imas/3.12.0* at least), and some number of added/removed source code files, it is likely that the compilations launched by the update scripts will fail the first time. If this is the case, start afresh in a new window, load the SOLPS-ITER environment as usual by means of the *setup.csh* file, and try again to compile. If it fails once more, then do a *"gmake clean"* first and compile again.

2. Changes to the transport equations:

Please be mindful that the changes to the transport equations described in this section will most likely cause some evolution of existing 3.0.5 solutions. How much will depend on the specifics of the cases at hand. The benchmarks and sample cases available on the SOLPS-ITER website all now have been recomputed using the 3.0.6 physics model.

a. New friction and thermal force expressions:

The parallel momentum balance equation has been revised, in particular the friction and thermal force terms (including associated energy sources and corrections to the parallel current expression), following work by Lisa Sytova and colleagues in St. Petersburg. It is activated by setting the switch "*b2sigp_style*" to "2" (which is the new default in 3.0.6), and the older version of the equation can be recovered with "*b2sigp_style*" equal to "1". This new description introduces two new flux limiters, for the thermal and friction forces. The consequences of the introduction of these new variables on the input files are detailed in section 3.b of this document below.

For details of the implementation of the new equations, please consult Appendices B.4 and C.2 of the manual. The new formulation of the equations applies for cases where the main ion species is also the lightest, and there is only one main hydrogenic species (if any) [i.e., no 50/50 DT mixtures], *and* where the potential equation is being solved. It is most relevant for cases where $Z_{\text{eff}} \sim 1.5$ or larger, and will likely lead to a redistribution of the impurities followed in the simulation. The code warns the user if the assumptions leading to the derivation of the revised parallel momentum balance equations are not satisfied.

A recently updated document providing a description of the recommended physics model equations as implemented in v.3.0.6 can be found in [*\\$SOLPSTOP/doc/B2solps5.2_equations_13.11.2017.pdf*](#).

b. Adjustments to flux conductivities:

The computation of the fluxes has been modified slightly to enforce that the conduction component is the same whether the *b2tfhe/hi/nb_mdf* switches are used or not (i.e. which treatment is chosen for removing the non-divergent part of the diamagnetic flows). This may affect plasma solutions in some cases. Also the flow calculations were changed in such a way as to provide the same conductive particle flows entering all the equations (momentum balance, heat balances, and particle balances).

c. Parallel velocity restrictions and instability control:

In v. 3.0.5, some cases displayed numerical instabilities in the parallel velocity (in particular for impurity species with a very low relative density). Three changes to the code have been implemented to address this issue. First, the pressure gradient term calculation has been modified to correct for cases with insufficient poloidal resolution. Second, a new switch "*b2npmo_ion_vlct_restrict*" (default "*0*") has been introduced. If set to "*1*", the parallel velocity is limited to Mach 3. Additionally, if set to "*2*", the $\mathbf{E} \times \mathbf{B}$ velocities (poloidal and radial) are limited to Mach 2. By default, the code will give a warning message providing a count of how many times, for each species, these velocity limits have been applied. A new switch "*b2npmo_vlct_diagno*" (default "*1*") allows to silence this output (when set to "*0*") or obtain full details (when set to "*2*", which can however produce lots of output). Third, the code now enforces that the amount of poloidal magnetic flux enclosed between two neighbouring radial surfaces is constant poloidally (see Section 3.a of this document below).

d. Currents assigned to ions:

Electric currents are defined as the difference between electron and ion particle flows when comparing them to ambipolar flow. In SOLPS5.x, it was assumed that all currents were carried by electrons, i.e. the full difference was assigned to the electron velocity. In SOLPS-ITER, this is no longer automatically the case. In v. 3.0.6, it is now possible to assign the currents due to ion-neutral friction, parallel viscosity, perpendicular viscosity, inertia, heat viscosity, and anomalous current to be carried by the hydrogenic ions instead of electrons (see manual sections C.1, C.4.2 and C.4.4 for the corresponding equations). This feature can only be used if the simulation contains a single hydrogenic ion species (to avoid double-counting of the currents) and the code will check for this. The recommended (default) model is to assign the radial currents (with the exception of the perpendicular viscosity current) to the ions, but not the poloidal currents. These settings can be modified by means of the switches "*b2tfnb_xcur*" (default "*0.0*", assigns the poloidal currents to ions), "*b2tfnb_ycur*" (default "*1.0*", assigns the radial currents to ions), "*b2tfnb_vis_q*" (default "*1.0*", assigns the heat viscosity current to ions) and "*b2tfnb_vis_per*" (default "*0.0*", assigns the perpendicular viscosity current to ions).

Importantly, the *fac_vis* array is now used as a space-variable multiplier for the three viscosity currents (parallel, perpendicular, and heat), while *facdrift* is used to multiply the ion inertia current.

The ion-neutral friction current *fchin* is also only activated if at least one of the $\mathbf{E} \times \mathbf{B}$ or diamagnetic drifts is turned on. A new switch '*b2tinnt_fchin_in_core*' allows for turning off this current in the core region of coupled runs (default), so as to avoid

this current being dominated by Monte-Carlo noise in regions where the neutral densities are very low and thereby causing convergence problems.

The calculation of the perpendicular viscosity current *fchvisper* has been revisited. It now uses its own solver routine *ma28copy3* with a separate copy of the solver arrays and common blocks. As a consequence, the default value of the “*b2ux7p_style*” switch has been reverted from “3” back to “2” and the *b2usp7_.F* routine has been removed, while a new *b2txvspr.F* routine is used to pre-compute the various terms involved.

e. Stochastic current and anomalous conductivity:

A new momentum source term has been added to correspond to the radial anomalous current being carried by ions. This is activated by the new switch “*b2sian_phm0*” (default value is “1.0”).

Additionally, there is a new optional stochastic current term that may be turned on by means of the “*b2tfhe_stochastic*” switch (“0.0” by default). See Section C.5.2 of the manual for a definition.

f. Separation of classical and anomalous viscosity terms:

A more accurate account of classical and anomalous viscosity terms and of the factor of 4/3 in the parallel balance momentum equation and terms connected with viscosity has been done. For example, instead of *cvsa*, it is the classical viscosity *cvsa_cl* that is used for all calls of to the subroutine *b2tvspa*. This change does affect simulations results to a small extent.

The calculation of sources *shivc* and *shiva* has been made more uniform and the contribution from fluid neutrals has been taken into account for cases run without Eirene.

3. Changes to the input files:

a. Enforcing constant poloidal magnetic flux:

The new default behaviour of **b2ag** (using the “*b2agdr_redef_pbs*” switch, set to “1” by default) is now to enforce a constant poloidal magnetic flux for each flux tube. The effect of this switch can be seen by the “sanity checks” at the end of **b2ag** indicating that *hx*, *hy*, and *pbs* have changed by something of order 1% or so. Please also remove any mention of the “*b2mndr_redef_pbs*” switch from your *b2mn.dat* files, if any, as this is now superseded by “*b2agdr_redef_pbs*” in *b2ag.dat* instead and the code will stop if the former switch is found.

In v. 3.0.6, the *b2fgmtry* file will now contain a new field corresponding to the value of *redef_pbs* that has been used to build that file, so it is known whether the poloidal flux correction has already been applied.

b. New flux limiters *cflal* and *cflab*:

As part of the new formulation for the parallel momentum balance equation (cf. Section 2.a above), two new flux limiters have been introduced, namely *cflal* for the thermo-electric coefficient (equivalent to *cflim(3)*), and *cflab* for the friction force (equivalent to *cflim(4)*). These are set in *b2ah.dat*, the optional **b2cmpt* block in *b2mn.dat*, and/or in *b2.transport.parameters*. See Sections C.3.3 and C.2.1 in the manual for the exact formulation of these new flux limits.

c. Avoiding Eirene restart effects:

In v. 3.0.5, restart effects were reported for some Eirene runs. These were identified to be related to the presence of BGK species and incorrect usage of the *fort.13* file. Eirene has been corrected to read and write *fort.13* in the correct sequence, and the restart effect is now avoided, provided the NFILE switch in the Eirene input file is set to at least 300 (check the Eirene manual for full details on the usage of NFILE). The *runs/Makefile* has also been modified to keep a copy of all SOLPS-ITER start-up files (with a *.i* extension) that are otherwise overwritten in the course of a simulation. Files *fort.13* and *fort.15* have also been added to the set of savefiles that are written out regularly for recovery after crashes, using the **b2co** utility. In v. 3.0.6, the **Uinp** program will provide, if BGK species are part of the Eirene model description, a consistent NFILE value in the Eirene input files.

d. Additional correspondence arrays between Eirene and B2.5 species:

As part of the adaptations required for the nitrogen chemistry package (see Section 4.a below), it has become necessary to distinguish between the number of atoms described in the Eirene input file (often denoted by “*natmi*”) and the number of isonuclear sequences followed by B2.5 (denoted by “*nspecies*”), to allow for the possibility that, on the Eirene side, an atomic species may be subdivided into several separate excitation states. As a consequence, three new optional arrays have been added to *b2.neutrals.parameters*, namely *LATMSCL*, *EB2ATCR* and *B2EATCR*, which respectively indicate to which B2.5 sequence each Eirene atom belongs, to which B2.5 species each Eirene atom maps, and to which Eirene atom each B2.5 species is linked. Complementarily, the meaning of *B2ESPCR* changes slightly, to refer to the isonuclear sequence number to which a B2.5 species belongs, instead of, previously, the index of the corresponding Eirene atom species. The code attempts to fill these arrays by default but users should check these definitions in cases where the mapping is not one-to-one. The “*b2eatcr*” array is now also accessible by **b2plot**.

There are also changes to the *j_h_at* and *l_h_mol(1:nmol)* arrays from *b2.user.parameters*, which, in order to handle cases with hydrogen isotope mixtures, have been given an extra dimension corresponding to the isotope mass, i.e.

$j_h_at(1:3)$ and $l_h_mol(1:nmol,1:3)$ where the 1:3 index runs through H/D/T isotopes. To maintain backward compatibility, it is still possible, for cases with only one hydrogen species, to specify the first element only of j_h_at and l_h_mol , meaning that the format of *b2.user.parameters* can stay unchanged, and the code will do any necessary re-assignment internally. It is also still possible to build a “red deuterium/blue deuterium” case (where one has two species for the same isotope, sometimes used for numerical tests), provided the j_h_at and l_h_mol arrays still are consistent with each other. **Uinp** and **b2yt** will now automatically create *b2.user.namelist(.2)* files according to the new format.

e. Boundary conditions:

After a careful review of the standard sheath boundary conditions (*BCENE/I=3*) by David Tskhakaya, the new recommended value for the *GAMMAI* factor is 1.0 instead of the traditional 5/3. As part of that review, the ion energy carried back to the plasma by reflected neutrals, when using the fluid neutral model, has been made proportional to the ion incident energy $E_i = T_i + Z_a \Delta\Phi$, where $\Delta\Phi$ is the sheath potential drop, instead of $(Z_a T_e + T_i)$.

The SPB sheath boundary conditions (*BCENE/I=15*) are now linked to the simultaneous use of *BCCON=14*, and the sign convention for the corresponding *ENIPAR* parameter has been flipped to match *BCENI=3*, i.e. a positive parameter corresponds to plasma flow directed towards the outside of the computational domain. (The boundary condition *BCENE=15* does not make use of the *ENEPAR* array.)

f. New default boundary arrays dimensions:

The *\$\$SOLPSTOP/modules/B2.5/src/include/DIMENSIONS.F* includes two new optional dimensions, *DEF_NBC* and *DEF_NISO*, corresponding to the maximum number of allowed boundary conditions (previous default was a hard-coded 20, now it is $6*(DEF_NCUT/2+1)$, i.e. 18 for the default parameter set) and the number of isolated regions (default value is unchanged at 10). Also, to match the ITER examples being provided and used for the SOLPS-GUI tutorials, the value of *DEF_NSRRFS* has been increased to 4.

g. Obsolete switches

The following input switches from *b2mn.dat* have been made obsolete:

- “*b2npht_style_shei*” and “*b2nph9_style_shei*”, meaning that ion-electron temperature equipartition source terms will no longer be applied in guard cells.

- “*b2tfnb_cfvma*”, meaning the total convective flow (that which takes into account the true diamagnetic flow) will be computed using the same (if any) anomalous pinch velocity as the other fluxes.
- “*b2tfnb_fnb_nodrift_style*”, which had value '1' by default when SPb's form of calculating the no-drift part of the particle fluxes was used. Now the *fna_nodrift* array is calculated as per the old treatment according to the zero value of switch “*b2tfnb_fnb_nodrift_style*” which has now been removed.
- “*b2tfhe_inert*” has been superseded by “*b2tiner_inert*”.
- “*b2npp7_style*” is no longer needed as the *b2usp7.F* and *b2usp7_.F* subroutines have been replaced by a single version of *b2usp7.F*.

h. New pressure correction speed-up scheme

A new speed-up scheme is now available, which acts on the pressure correction terms in the core region. A full description of the scheme is available in *\$SOLPSTOP/doc/Pressure_correction_speed-up.pdf*. It is governed by two new switches, to be found in *b2.numerics.parameters*, namely *corr_core_dt* for the temperature term and *corr_core_dn(is)* for the density term from species *is*.

4. New features:

a. IDS support:

SOLPS-ITER now has some utilities meant to produce/read/archive Interface Data Structures (IDSs) for use within IMAS (Integrated Modelling and Analysis Suite), in particular so as to establish a database of reference cases and integrated modelling workflows. The IDSs involved in SOLPS-ITER are *edge_profiles*, *edge_sources*, *edge_transport*, and *transport_solver_numerics*. The exact content of these IDSs may still evolve, but some skeleton capabilities are already available. The new executables provided are:

- *cpo2ids.py* : A Python utility that converts ITM edge CPO to IMAS *edge_profiles* IDS. It opens an existing CPO, reads it and writes the data to IDS.
- *b2_ual_write* : Reads the plasma grid geometry (including Nodes and Cells grid subsets in native mode) and plasma state (electron density, electron temperature, ion temperature) and writes it to IDS database.
- *b2_ual_write_gsl* : Reads the plasma grid geometry (including Nodes and Cells grid subsets) and plasma state (electron density, electron temperature, ion temperature) and writes it to IDS database with the use of IMAS GGD Grid Service Library routines.
- *b2_ual_write_b2mod* : Reads the plasma grid geometry (full geometry descriptions of all available grid subsets) and plasma state (electron density) and writes it to IDS database with the use of *b2mod*

inside *modules/* and *ids/* that utilize IMAS GGD Grid Service Library routines.

The latter is the most advanced writer and recommended for use. Additional information can be found in the document:

\$SOLPSTOP/doc/IDS_visualization_DejanPenkoMSc.pdf, and among the “Various other presentations” listed on the SOLPS-ITER website, one can find, embedded in the “*IMEG2017_IMAS_Visualisation*” presentation, some Youtube video demos using these tools within the SOLPS-GUI environment.

b. Nitrogen chemistry:

As part of an ITER contract with CNRS-LSPM, a new comprehensive nitrogen chemistry database has been added to the SOLPS-ITER distribution. This database can be found, in the form of a LaTeX file readable by Eirene named *AMMONX*. There are four new AMDS file pairs associated with this database:

- *N2_only.am[do]*, which contains reactions related to N₂ molecule break-up;
- *Ammonia_add.am[do]*, which contains the reactions that need to be added to an existing hydrogen model to also include N₂ molecule break-up and ammonia formation;
- *Ammonia.am[do]*, which contains a reduced stand-alone chemistry model for an N-H plasma;
- *Ammonia_metastables.am[do]*, which contains a fuller stand-alone chemistry model for an N-H plasma including N and N₂ metastable states.

Inclusion of any of these AMDS files can be done in addition to already existing ones. Although **Uinp** will remove exact duplicates, users should still take care that they do not include two different descriptions of the same reaction pathway in two separate AMDS files. **Uinp** will also check whether the declared numbers of atom, molecule and test ion species (*DEF_NATM*, *DEF_NMOL* and *DEF_NION*, respectively) as well as the number of reactions (*DEF_NREAC*) are set to sufficiently large values in the *DIMENSIONS.F* file, and return an error message if this is not the case.

Full details are provided in a report found as *\$SOLPSTOP/doc/Ammonia_report.pdf*. Concerning ammonia formation, many of the pathways involve surface reactions for which the capability has not yet been coded into Eirene, and so should only be used for experimental purposes.

c. Eirene source averaging scheme:

A new scheme based on following an averaged solution, provided by KU-Leuven and Jülich colleagues, has now been implemented. A short description is provided in the Manual, Sections 5.4 and 5.5, and a more complete report is available as

\$SOLPSTOP/doc/SOLPS-ITER_Eirene_averaging.pdf. Please refer to that document for full details.

d. New storage scheme for reference cases:

There is now a new additional way to store, share, and publish SOLPS-ITER cases. A series of MD5 hashes is stored in the *\$SOLPSTOP/runs/examples* directory. Each hash points to a full run directory stored on the IO cluster at */work/imas/shared/boundary/solps-iter/assets/MD5*, where it can be retrieved. Fuller information is available in the *\$SOLPSTOP/runs/examples/README.md* file. Of course, the set of cases stored can and will be expanded following user's suggestions and contributions in the future.

e. DivGeo3:

As part of the SOLPS-ITER 3.0.6, a new version of DivGeo is available (DivGeo3), in the *divgeo3* branch of the DIVGEO repository. This version has been built and provided by the St. Petersburg group for environments where the Motif library is unavailable and the standard version of DivGeo cannot be compiled (such as Windows). The recommended and default DivGeo version for SOLPS-ITER remains, for the time being, DivGeo2.

f. Looping job submission:

If you are interested in automatically relaunching runs after they finish, Ilya Senichenkov from St. Petersburg has provided some sample new submission scripts *QSUB.itm_loop* and *QSUB.itm_coupled_loop* for this purpose.

g. Python3 support:

Thanks to the efforts from our Ljubljana colleagues, the analysis scripts are now compatible with Python 3.

h. Miscellaneous new switches:

Several new switches have been introduced, among which are:

- *"b2mndr_old_style"*, which allows to create output in the old SOLPS4.x style.
- *"b2tfhe_prl_cur"*, default *"1.0"*, a multiplier to the parallel current.

5. Changes to the output:

a. Additions to *fort.46* file:

New fields have been added to the *fort.46* file. These are the triangle volumes, and the unit vector components for the radial and poloidal directions ($P[UV][XY]$) computed at the centroid of each of the Eirene grid triangles which are also part of

the B2.5 computational grid. The format header corresponding to the presence of these new data fields is “20170930”. The new quantities can also be plotted by **b2plot**.

b. New definition for total heat flux f_{ht} :

The definition of “total energy flux” diagnostic quantities has been changed to use the 5/2 formalism. This will affect files such as *ft3dr.last10*, etc... as well as the fluxes saved in MDSplus trees. The new definition is:

$$q_t = \frac{5}{2}\Gamma_e T_e + \frac{5}{2}\Gamma_i T_i - j\varphi + \sum_a \Gamma_a \left(\frac{1}{2} m_a u_a^2 + E_a^i \right) + \text{conductive terms}$$

c. New location for $iout$ *.dat files:

The new default location for the *.dat files that are printed out by the code when “*_iout” switches such as “b2wdat_iout” are activated, is an *output/* subdirectory created as part of the *b2mn.exe.dir/* temporary directory where the simulation is taking place. A new setting of “4” for “b2wdat_iout” gives a restricted set of “essential” output files. A description of possible output files is given in *\$SOLPSTOP/doc/Output_description.pdf* (provided by Lisa Kaveeva).

d. New treatment of tracing output files:

i. Change of header for *residuals.trc* and *user.trc* files:

The header for the *residuals.trc* file contained a mistake. The third integer on the first line needs to correspond to the number of ionic species followed by B2.5, not the total number of species. If the header is not changed and the code attempts to read the file, it will return an error and not write new data in the file. This should be changed by hand if you wish to keep that file, or the file should be removed.

A mismatch in formats between *user.trc* and *user.trc.n.01* (where the first was using *i8* and the second *i12* formats) has been removed, and the format is now *i8* throughout. The *user.trc.n.01* file, if any, is now only used internally and overwrites the *user.trc* file at the end of the run. This was done to simplify the appending feature in the next subsection.

ii. Appending files:

When continuing the time sequence of a run, by means of “b2mndr_stim” switch set to a negative number, new tracing data for the *.nc and *tracing/*.trc* files will be appended to the already existing files. Please bear in mind that, above a certain critical file size, NetCDF can choke and cause the program to crash if you attempt to append to a *.nc file that is too big.

Two new scripts, *save_optional_files*, and *move_optional_files*, are used by the runs Makefile to manipulate these files automatically. There is also a script, *recover_initial_files*, that can be used to reset the tracing files to their state before the run started, for example if the run crashed early and you wish to recover exactly the same starting point.

When running the *b2co* program for crash recovery of a run, an attempt will be made to recover the tracing data up to the time point specified in the plasma save file and rewrite the tracing and NetCDF data files so that they are synchronized with the recovered plasma state.

It was also possible to run *b2co* *while* a simulation was running in order to obtain a temporary *b2fstate* file (to be used for inline analysis/plotting). In order to still do this, but without overwriting the NetCDF and tracing data files, *b2co* **MUST** be run with a new “-b” option, i.e. “*b2run -b b2co*”.

iii. **New file content:**

The *integral.trc* file now contains additional data:

- “*K_en_ri_*”, total kinetic energy content per region;
- “*N_a_tot_*”, total atom content;
- “*N_a_reg_*”, atom content per region;
- “*Na_stra_*”, total atom content due to each Eirene stratum;
- “*sna_stra_*”, total Eirene particle sources per stratum.

Another new file is created, *intshrt.trc*, which contains a shorter set of integrals also found in *integral.trc*. These are the total plasma energy content (by region), total particle content (per region and isonuclear sequence) and total neutral particle content (per region and isonuclear sequence).

iv. **Choice of which tracing files are produced :**

The *b2.user.parameters* namelist now contains a new array of 10 logicals, named *FILEDATA*, which allows for turning on or off the production of each optional output file in the *tracing/* directory. This array already existed, but was internally hard-coded. By default, all array elements are set to *.true.*, meaning the file will be produced. However, if the file is to be appended and a reading error occurs when it is originally opened, then the setting will be changed to *.false.* for that file. Similarly, the *user.trc* file will not be produced if there is no *b2.user.parameters* file present in the run directory. For the balance files *blnn/e/m*, if the tracing file is not present, the output will be redirected to the run log file. The correspondences are:

1. *FILEDATA(1)* → *intshrt.trc*
2. *FILEDATA(2)* → *test.trc*

3. *FILEDATA(3) → residuals.trc*
4. *FILEDATA(4) → sources.trc*
5. *FILEDATA(5) → blnn.trc*
6. *FILEDATA(6) → blne.trc*
7. *FILEDATA(7) → integral.trc*
8. *FILEDATA(8) → user.trc*
9. *FILEDATA(9) → blnm.trc*
10. *FILEDATA(10) → sepdata.trc*

e. Eirene spectra:

When asking for energy spectra incident on surfaces in block 10F of the Eirene input file, it is now possible to ask for the incident energy spectra to be binned logarithmically instead of linearly, by setting NSPS to a negative value.

f. New averaging option for *balance.nc* file:

There is now a switch capability (provided by David Moulton) to obtain the power, particle and momentum balance arrays averaged over an entire run, by means of the switch "*balance_average*". If set to "1" (default value is "0"), the balance arrays in the *balance.nc* file will be averaged over the entire duration of the run, thereby smoothing out eventual statistical noise from Eirene Monte-Carlo sources.

g. Other additional b2plot output:

In the wall loading output files, obtained by the "*1000 wlld*" command from **b2plot**, the heat load due to plasma particles is further decomposed into its thermal+kinetic component "*Whtpl*" and the heat load due to the recombination energy of the incident particle flux "*Wptpl*".

Other new **b2plot** quantities include:

- "*fluce*", Luciani flux limit factor applied to the electron thermal conductivity,
- "*fluci*", Luciani flux limit factor applied to the ion thermal conductivity,
- "*flucv*", Luciani flux limit factor applied to the ion viscosity,
- "*flucs*", Luciani flux limit factor applied to the electric conductivity,
- "*fluca*", Luciani flux limit factor applied to the thermo-electric coefficient,
- "*flke*", classical flux limit factor applied to the electron thermal conductivity,
- "*flki*", classical flux limit factor applied to the ion thermal conductivity,
- "*flal*", classical flux limit factor applied to the thermo-electric coefficient,
- "*flalc*", same as "*flal*", but computed at cell centers.

Concerning the quantities above, remember that the Luciani flux limits apply on closed field lines, while the "classical" flux limits apply on open field lines. Only one or the other is ever applied in any particular location. If you are interested in the

older formulation of flux limits which were applied to fluxes as opposed to transport coefficients, these are still available as “*fle*”, “*fli*” and “*flvisc*”.

6. Getting a case restarted with v. 3.0.6:

As a consequence of all the changes above, in order for users to take full advantage of the new 3.0.6 features, users wishing to continue their older runs should proceed in the fashion detailed below (and in the same order). Of course, if converting older cases or building new cases from scratch, the **Uinp**, **b2yt** and **b2sxd** programs have been modified to produce input files already compatible with v. 3.0.6, which should not preclude a visual inspection of all relevant input files, as is best practice.

1. If your cases include a very large number of boundaries and/or isolated regions (more than 18 and 10, respectively), add or modify the values of *DEF_NBC* and/or *DEF_NISO* in your local copy of the *DIMENSIONS.F* file (see Section 3.f above), then recompile your code.
2. If you have a case using Nitrogen injection, you can now choose, for instance, to inject N₂ molecules instead of N atoms. To do so, go back to your DivGeo model, modify your gas puff species (Variables | Gas puff | Gas species | “N2”), and add an AMDS reference (Variables | Reference to AMDS | “N2_only.amo”) to the ammonia database, then re-run **Uinp**, after clearing your older *b2.*.parameters* files to allow for them to be rewritten. The **Uinp** program has also been modified to handle some of the nitrogen reactions that produce 3 secondaries and/or are neutral-neutral reactions that require the inclusion of BGK species in the Eirene input files, all of which are automatically handled.
3. Re-run **b2ag** (in your *baserun/* directory, after checking that you have a valid *b2ag.dat* file and deleting *b2ag.prt*) to obtain a new *b2fgmtry* file where the poloidal flux constancy along flux surfaces has been enforced.
4. Re-run **b2ah** (in your *baserun/* directory), after adding your desired values for *cflal* and *cflab* (recommended default values are 0.5 for both variables) to obtain a new *b2fpardf* file. These variables are given at the end of the *b2ah.dat* file, which you should modify to read something along the lines of:

*cflim	cflme	cflmi	cflmv	cflal	cflab	...
'-1'	0.15	0.15	0.5	0.5	0.5	0 0 0

You should also check your *b2mn.dat* files for similar lines in the **b2cmpt* block (near the beginning of the file), and modify accordingly as well.
5. Re-run **b2ai** (in your *baserun/* directory), to update your *b2fstati* file. Because you have done this, remember to do “touch b2fstati” afterwards in each of the case directories on the parallel level to your baserun/ directory!!!
6. Check your Eirene input files to make sure that the value of NFILE matches your intended use of *fort.1[0-5]* files, in particular that you read and write

fort.13 if using BGK species and read and write *fort.15* if you use a time stratum. Refer to the Eirene manual if you are unsure of which setting to use.

7. Decide if you want to activate the ion parallel velocity restrictions in *b2mn.dat* with the switch "*b2npmo_ion_vlct_restrict*" and the desired level of output with "*b2npmo_vlct_diagno*".
8. If you do not wish to change the parallel momentum balance formulation, or your case does not meet the new formulation derivation assumptions, set "*b2sigp_style*" to "1" in your *b2mn.dat* file.
9. If running cases with drifts, decide if you want to assign poloidal and radial currents (and which currents) to be carried by the hydrogenic ions instead of the electrons, by means of "*b2tfnb_xcur*", "*b2tfnb_ycur*", "*b2tfnb_vis_q*", and "*b2tfnb_vis_per*". Recommended values are 0.0, 1.0, 1.0 and 0.0, respectively. *xcur* and *ycur* should only be used in cases which contain one and only one hydrogenic ion species. For cases without drifts but with some anomalous conductivity, you must also decide how that current is assigned and set "*b2tfnb_xcur*" and "*b2tfnb_ycur*" accordingly. You should also change the value of the "*b2ux7p_style*" switch, if set to "3", to "2".
10. If using a *b2.transport.parameters* file, it is recommended to include in it the settings you wish to have for the *CFLAL* and *CFLAB* variables.
11. If using a *b2.boundary.parameters* file, it is recommended to change the value of *GAMMAI* from 5/3 to 1.0. If using *BCENE/I=15*, make sure the density boundary condition on the same boundary is set to *BCCON=14* for all ions, and check the sign of the *EN[EI]PAR* coefficients, which should now always be positive.
12. If running a case with more than one hydrogen species, make sure that the *j_h_at* and *l_h_mol* assignments in *b2.user.parameters* point to the isotopes you want to use. *j_h_at(1)* is H, *j_h_at(2)* is D, and *j_h_at(3)* is T.
13. If you wish to re-use your *tracing/residuals.trc* file, change the third integer in its top header line to correspond to the number of ionic (charged) species in your simulation, instead of the total number of species.
14. The new scheme allowing for appending NetCDF files requires the format and number of variables in each file to be the same from one run to the next. If this is not the case, the code will return an error and stop at the first attempt in writing the NetCDF files. To avoid this error, either move the older *.nc to a different name or set '*b2mndr_stim*' to zero or to the actual current plasma time, so that the *.nc files are restarted anew with the correct format and variables list. This is particularly true if you are turning on the Eirene source averaging scheme (see point 15 below), which introduces new variables in *b2time.nc*.
15. If you wish to use the Eirene averaging scheme, follow the procedure outlined in *\$SOLPSTOP/doc/SOLPS-ITER_Eirene_averaging.pdf*.

7. Additional notes for developers:

- i. A new logical variable was introduced: *on_closed_surface(ix,iy)*, which is *.true.* for cells on closed flux surfaces (i.e. in core regions, in annular cases, and in divertor island regions for stellarator geometries), and *.false.* otherwise. Checking against this variable should be used instead of checking if the region number is 1 or 5, as the latter test will misidentify slab cases with a single region as core regions.
- ii. Because it can no longer be assumed that *natmi* (the number of atomic species in the Eirene input file) and *nspecies* (the number of isonuclear sequences being followed by the B2.5 plasma model) are equal, care must be taken when manipulating variables that are so dimensioned. Please make use of the *latmscl*, *b2eatcr*, *b2espcr*, *eb2sPCR* and *eb2atcr* relationship arrays to properly map between Eirene atoms, B2.5 (fluid neutral and ion) species, and B2.5 isonuclear sequences.
- iii. Two new compiler options (*-ffpe-trap=denormal -ftrapv*) have been added to the debugging flags with using the gfortran compiler. This allows the debugger to catch both the use of uninitialized variables after they have been passed as array arguments and integer overflows, which was not possible with the previous options. This new flag may also catch other minor code mistakes that have not yet been detected. Please report these back if and when you encounter such issues.
- iv. The computation of the currents have been split into several routines, such as *b2tpeSch*, *b2tstch*, *b2tvSq*, *b2tvSpa*, *b2tvSpr*, each specializing into a single current contribution. It also makes the coding in *b2tfhe_* and *b2tfch* much simplified.